

Occupational Safety and Health (Classification, Labelling and Safety Data Sheet of Hazardous Chemicals) Regulations 2013

**SECTION 1: IDENTIFICATION OF THE SUBSTANCE/MIXTURE AND OF THE COMPANY/UNDERTAKING**
**Product Identifier**

|                             |                                |
|-----------------------------|--------------------------------|
| <b>Perihal Produk:</b>      | <b><u>Manganese pieces</u></b> |
| <b>Product Description:</b> | <b><u>Manganese pieces</u></b> |
| <b>Cat No. :</b>            | 96027                          |
| <b>CAS No</b>               | 7439-96-5                      |
| <b>Molecular Formula</b>    | Mn                             |

**Relevant identified uses of the substance or mixture and uses advised against**

|                             |                          |
|-----------------------------|--------------------------|
| <b>Recommended Use</b>      | Laboratory chemicals.    |
| <b>Uses advised against</b> | No Information available |

|                |                                                                                                                                                                                                                                                       |
|----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Company</b> | Thermo Fisher Scientific Fisher Scientific (M) Sdn Bhd<br>Hap Seng Business Park, Lot 01-03, 01-04 Aras 1 Unity Square,<br>No 12, Persiaran Perusahaan, Seksyen 23, 40300 Shah Alam,<br>Selangor Darul Ehsan, Malaysia.<br>Main line: +60 3-5525 7888 |
|----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

**Supplier**

|                       |                             |
|-----------------------|-----------------------------|
| <b>E-mail address</b> | Enquiry.my@thermofisher.com |
|-----------------------|-----------------------------|

|                                   |                                                                                                                                         |
|-----------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|
| <b>Emergency Telephone Number</b> | Tel: +03-5525 7888<br>CHEMTREC Malaysia <b>1-800-815-308</b> (Malay)<br>CHEMTREC Malaysia (Kuala Lumpur) <b>+(60)-327884561</b> (Malay) |
|-----------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|

**SECTION 2: HAZARDS IDENTIFICATION**
**Classification of the substance or mixture****Label Elements**
**Hazard Statements**
**Precautionary Statements**
**Storage**

P403 - Store in a well-ventilated place

**Disposal**

P501 - Dispose of contents/ container to an approved waste disposal plant

**Other Hazards**

# SAFETY DATA SHEET

Manganese pieces

Revision Date 29-Mar-2025

This product does not contain any known or suspected endocrine disruptors

## SECTION 3: COMPOSITION/INFORMATION ON INGREDIENTS

| Component | CAS No    | Weight % |
|-----------|-----------|----------|
| Manganese | 7439-96-5 | <=100    |

## SECTION 4: FIRST AID MEASURES

### Description of first aid measures

#### Eye Contact

Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Get medical attention.

#### Skin Contact

Wash off immediately with plenty of water for at least 15 minutes. Get medical attention immediately if symptoms occur.

#### Ingestion

Clean mouth with water and drink afterwards plenty of water. Get medical attention if symptoms occur.

#### Inhalation

Remove to fresh air. Get medical attention immediately if symptoms occur.

#### Self-Protection of the First Aider

No special precautions required.

### Most important symptoms and effects, both acute and delayed

None reasonably foreseeable.

### Indication of any immediate medical attention and special treatment needed

#### Notes to Physician

Treat symptomatically.

## SECTION 5: FIREFIGHTING MEASURES

### Extinguishing media

#### Suitable Extinguishing Media

approved class D extinguishers.

#### Extinguishing media which must not be used for safety reasons

Water may be ineffective.

### Special hazards arising from the substance or mixture

Thermal decomposition can lead to release of irritating gases and vapors.

#### Hazardous Combustion Products

Manganese oxides.

### Advice for fire-fighters

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear.

## SECTION 6: ACCIDENTAL RELEASE MEASURES

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# SAFETY DATA SHEET

Manganese pieces

Revision Date 29-Mar-2025

## Personal Precautions, Protective Equipment and Emergency Procedures

Ensure adequate ventilation. Use personal protective equipment as required. Avoid dust formation. No special precautions required.

## Environmental precautions

Should not be released into the environment. See Section 12 for additional Ecological Information. Do not allow material to contaminate ground water system. Do not flush into surface water or sanitary sewer system.

## Methods and Material for Containment and Cleaning Up

Sweep up and shovel into suitable containers for disposal. Avoid dust formation. Pick up and transfer to properly labelled containers.

## Reference to Other Sections

Refer to protective measures listed in Sections 8 and 13.

## SECTION 7: HANDLING AND STORAGE

### Precautions for Safe Handling

Avoid contact with skin, eyes or clothing. Avoid ingestion and inhalation. Wear personal protective equipment/face protection. Ensure adequate ventilation. Avoid dust formation.

### Conditions for Safe Storage, Including any Incompatibilities

Keep in a dry place. Keep away from acids.

### Specific End Uses

Use in laboratories.

## SECTION 8: EXPOSURE CONTROLS/PERSONAL PROTECTION

### Control Parameters

| Component | Malaysia | ACGIH TLV                                                 | OSHA PEL                                                                                                                                            |
|-----------|----------|-----------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|
| Manganese |          | TWA: 0.02 mg/m <sup>3</sup><br>TWA: 0.1 mg/m <sup>3</sup> | (Vacated) TWA: 1 mg/m <sup>3</sup><br>Ceiling: 5 mg/m <sup>3</sup><br>(Vacated) STEL: 3 mg/m <sup>3</sup><br>(Vacated) Ceiling: 5 mg/m <sup>3</sup> |

| Component | European Union                                                   | The United Kingdom                                                                                                                               | Germany                                                                                                                                                                                                                                                                                                     |
|-----------|------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Manganese | TWA: 0.2 mg/m <sup>3</sup> (8h) TWA: 0.05 mg/m <sup>3</sup> (8h) | STEL: 0.6 mg/m <sup>3</sup> 15 min<br>STEL: 0.15 mg/m <sup>3</sup> 15 min<br>TWA: 0.2 mg/m <sup>3</sup> 8 hr<br>TWA: 0.05 mg/m <sup>3</sup> 8 hr | TWA: 0.2 mg/m <sup>3</sup> (8 Stunden). AGW - exposure factor 8<br>TWA: 0.02 mg/m <sup>3</sup> (8 Stunden). AGW - exposure factor 8<br>TWA: 0.2 mg/m <sup>3</sup> (8 Stunden). MAK<br>TWA: 0.02 mg/m <sup>3</sup> (8 Stunden). MAK<br>Höhepunkt: 1.6 mg/m <sup>3</sup><br>Höhepunkt: 0.16 mg/m <sup>3</sup> |

### Exposure Controls

#### Engineering Measures

None under normal use conditions.

# SAFETY DATA SHEET

Manganese pieces

Revision Date 29-Mar-2025

## Personal protective equipment

|                          |                                                    |
|--------------------------|----------------------------------------------------|
| Eye Protection           | Wear safety glasses with side shields (or goggles) |
| Hand Protection          | No special protective equipment required           |
| Skin and body protection | Long sleeved clothing                              |

|                          |                                                               |
|--------------------------|---------------------------------------------------------------|
| Respiratory Protection   | No protective equipment is needed under normal use conditions |
| Recommended Filter type: | Particle filter                                               |

|                         |                                                                       |
|-------------------------|-----------------------------------------------------------------------|
| <u>Hygiene Measures</u> | Handle in accordance with good industrial hygiene and safety practice |
|-------------------------|-----------------------------------------------------------------------|

|                                        |                          |
|----------------------------------------|--------------------------|
| <u>Environmental exposure controls</u> | No information available |
|----------------------------------------|--------------------------|

## SECTION 9: PHYSICAL AND CHEMICAL PROPERTIES

### Information on basic physical and chemical properties

|                |                          |
|----------------|--------------------------|
| Appearance     | Grey                     |
| Physical State | Solid disc Target        |
| Odor           | Odorless                 |
| Odor Threshold | No data available        |
| pH             | No information available |

|                     |                          |                                   |
|---------------------|--------------------------|-----------------------------------|
| Melting Point/Range | 1245 °C / 2273 °F        |                                   |
| Softening Point     | No data available        |                                   |
| Boiling Point/Range | 2150 °C / 3902 °F        |                                   |
| Flash Point         | No information available | Method - No information available |

|                          |                          |       |
|--------------------------|--------------------------|-------|
| Evaporation Rate         | Not applicable           | Solid |
| Flammability (solid,gas) | No information available |       |
| Explosion Limits         | No data available        |       |

|                              |                          |       |
|------------------------------|--------------------------|-------|
| Vapor Pressure               | 23 hPa @ 20 °C           |       |
| Vapor Density                | Not applicable           | Solid |
| Specific Gravity / Density   | No data available        |       |
| Bulk Density                 | 7.3 g/cm3                |       |
| Water Solubility             | Insoluble in water       |       |
| Solubility in other solvents | No information available |       |

### Partition Coefficient (n-octanol/water)

|                           |                          |       |
|---------------------------|--------------------------|-------|
| Autoignition Temperature  | No data available        |       |
| Decomposition Temperature | No data available        |       |
| Viscosity                 | Not applicable           | Solid |
| Explosive Properties      | No information available |       |
| Oxidizing Properties      | No information available |       |

|                   |    |
|-------------------|----|
| Molecular Formula | Mn |
|-------------------|----|

# SAFETY DATA SHEET

Manganese pieces

Revision Date 29-Mar-2025

## SECTION 10: STABILITY AND REACTIVITY

### Reactivity

None known, based on information available.

### Chemical Stability

Stable under normal conditions.

### Possibility of Hazardous Reactions

#### **Hazardous Polymerization Hazardous Reactions**

No information available.  
None under normal processing.

### Conditions to Avoid

None known.

### Incompatible Materials

Acids.

### Hazardous Decomposition Products

Manganese oxides.

## SECTION 11: TOXICOLOGICAL INFORMATION

### Information on Toxicological Effects

#### **Product Information**

##### **(a) acute toxicity;**

**Oral**

Based on available data, the classification criteria are not met

**Dermal**

No data available

**Inhalation**

No data available

| Component | LD50 Oral             | LD50 Dermal | LC50 Inhalation              |
|-----------|-----------------------|-------------|------------------------------|
| Manganese | LD50 = 9 g/kg ( Rat ) | -           | LC50 > 5.14 mg/L ( Rat ) 4 h |

##### **(b) skin corrosion/irritation;**

No data available

##### **(c) serious eye damage/irritation;**

No data available

##### **(d) respiratory or skin sensitization;**

**Respiratory**

No data available

**Skin**

No data available

##### **(e) germ cell mutagenicity;**

No data available

# SAFETY DATA SHEET

Manganese pieces

Revision Date 29-Mar-2025

(f) carcinogenicity; No data available  
There are no known carcinogenic chemicals in this product

(g) reproductive toxicity; No data available

(h) STOT-single exposure; No data available

(i) STOT-repeated exposure; No data available

Target Organs No information available.

(j) aspiration hazard; Not applicable  
Solid

Symptoms / effects, both acute and delayed No information available.

Endocrine Disrupting Properties Assess endocrine disrupting properties for human health. This product does not contain any known or suspected endocrine disruptors.

## SECTION 12: ECOLOGICAL INFORMATION

Ecotoxicity effects May cause long-term adverse effects in the environment. Do not allow material to contaminate ground water system.

| Component | Freshwater Fish                                               | Water Flea | Freshwater Algae | Microtox |
|-----------|---------------------------------------------------------------|------------|------------------|----------|
| Manganese | LC50: > 3.6 mg/L, 96h<br>semi-static<br>(Oncorhynchus mykiss) |            |                  |          |

Persistence and degradability Product contains heavy metals. Discharge into the environment must be avoided. Special pre-treatment is necessary  
Insoluble in water, May persist.  
Not relevant for inorganic substances.  
Contains substances known to be hazardous to the environment or not degradable in waste water treatment plants.

**Persistence**  
**Degradability**  
**Degradation in sewage treatment plant**

Bioaccumulative potential May have some potential to bioaccumulate; Product has a high potential to bioconcentrate

Mobility in soil Spillage unlikely to penetrate soil. Is not likely mobile in the environment due its low water solubility.

Endocrine Disruptor Information This product does not contain any known or suspected endocrine disruptors

Other adverse effects No information available

# SAFETY DATA SHEET

Manganese pieces

Revision Date 29-Mar-2025

## SECTION 13: DISPOSAL CONSIDERATIONS

### Waste treatment methods

**Waste from Residues/Unused Products**

Dispose of in accordance with local regulations

**Contaminated Packaging**

Empty containers should be taken to an approved waste handling site for recycling or disposal

## SECTION 14: TRANSPORT INFORMATION

IMDG/IMO

Not regulated

Road and Rail Transport

Not regulated

IATA

Not regulated

**Special Precautions for User**

No special precautions required

## SECTION 15: REGULATORY INFORMATION

### Safety, health and environmental regulations/legislation specific for the substance or mixture

**International Inventories**

X = listed

| Component | EINECS    | TSCA | DSL | PICCS | ENCS | ISHL | IECSC | AICS | KECL     |
|-----------|-----------|------|-----|-------|------|------|-------|------|----------|
| Manganese | 231-105-1 | X    | X   | X     | X    |      | X     | X    | KE-22999 |

### National Regulations

**Persistent Organic Pollutant**  
**Ozone Depletion Potential**

This product does not contain any known or suspected substance  
This product does not contain any known or suspected substance

## SECTION 16: OTHER INFORMATION

### Legend

**CAS** - Chemical Abstracts Service

**EINECS/ELINCS** - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

**PICCS** - Philippines Inventory of Chemicals and Chemical Substances

**IECSC** - Chinese Inventory of Existing Chemical Substances

**KECL** - Korean Existing and Evaluated Chemical Substances

**TSCA** - United States Toxic Substances Control Act Section 8(b) Inventory

**DSL/NDL** - Canadian Domestic Substances List/Non-Domestic Substances List

**ENCS** - Japanese Existing and New Chemical Substances

**AICS** - Australian Inventory of Chemical Substances

**NZIoC** - New Zealand Inventory of Chemicals

**WEL** - Workplace Exposure Limit

**ACGIH** - American Conference of Governmental Industrial Hygienists

**TWA** - Time Weighted Average

**IARC** - International Agency for Research on Cancer

# SAFETY DATA SHEET

Manganese pieces

Revision Date 29-Mar-2025

**RPE** - Respiratory Protective Equipment  
**LC50** - Lethal Concentration 50%  
**POW** - Partition coefficient Octanol:Water

**LD50** - Lethal Dose 50%  
**EC50** - Effective Concentration 50%

**ADR** - European Agreement Concerning the International Carriage of Dangerous Goods by Road  
**IMO/IMDG** - International Maritime Organization/International Maritime Dangerous Goods Code  
**OECD** - Organisation for Economic Co-operation and Development  
**BCF** - Bioconcentration factor

**ICAO/IATA** - International Civil Aviation Organization/International Air Transport Association  
**MARPOL** - International Convention for the Prevention of Pollution from Ships  
**ATE** - Acute Toxicity Estimate  
**VOC** - (Volatile Organic Compound)

## Key literature references and sources for data

<https://echa.europa.eu/information-on-chemicals>  
Suppliers safety data sheet, Chemadvisor - LOLI, Merck index, RTECS

|                         |                                             |
|-------------------------|---------------------------------------------|
| <b>Prepared By</b>      | Health, Safety and Environmental Department |
| <b>Revision Date</b>    | 29-Mar-2025                                 |
| <b>Revision Summary</b> | Not applicable.                             |

**In accordance with local and national regulations: Occupational Safety and Health (Classification, Labelling and Safety Data Sheet of Hazardous Chemicals) Regulations 2013**

## Disclaimer

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of Safety Data Sheet**